Titanic Dataset – Exploratory Data Analysis Report

Project: Task 5 – Exploratory Data Analysis (EDA)
Dataset: Titanic Dataset
Tools Used: Python, Pandas, NumPy, Matplotlib and Seaborn

1. Introduction

Exploratory Data Analysis (EDA) is the process of examining and visualizing data to identify patterns, trends, relationships and missing values.

In this project, EDA is performed on the Titanic passenger dataset to understand passenger information and survival patterns.

2. Objective

- Understand the structure of the Titanic dataset.
- Identify missing values and handle them.
- Perform statistical analysis.
- Create different visualizations.
- Identify relationships between passenger features.
- Summarize important findings.

3. Tools and Technologies

- **Python** – Programming language.
- **Pandas** – Data manipulation and analysis.
- **NumPy** – Numerical operations.
- **Matplotlib** – Data visualization.
- **Seaborn** – Statistical visualization.
- **Google Colab** – Running Python code.

4. Data Exploration

The dataset contains information about Titanic passengers, including age, gender, passenger class, ticket fare and survival status. The following functions were used:

- `head()` – Displays the first five rows.
- `info()` – Displays dataset information.
- `describe()` – Provides statistical summaries.
- `value_counts()` – Counts categorical values.

- `isnull().sum()` – Identifies missing values.

5. Data Cleaning

Missing values were identified using Pandas. The missing age values were replaced with the median, while missing embarkation values were replaced with the mode. A separate copy of the original dataset was used for cleaning.

6. Data Visualization

The following visualizations were created:

1. **Survival Count Plot** – Compares survivors and non-survivors.
2. **Gender Distribution** – Displays male and female passenger counts.
3. **Age Histogram** – Shows passenger age distribution.
4. **Gender vs Survival** – Compares survival across genders.
5. **Passenger Class vs Survival** – Examines survival across passenger classes.
6. **Age Boxplot** – Compares age distributions.
7. **Scatterplot** – Explores the relationship between age and fare.
8. **Correlation Heatmap** – Displays correlations between numerical variables.
9. **Pairplot** – Shows relationships between multiple numerical features.

7. Key Findings

- More passengers did not survive than survived.
- Female passengers had a higher survival rate than male passengers.
- First-class passengers generally had better survival chances.
- Passenger ages showed different distributions.
- The dataset contained missing values.
- Some numerical variables showed relationships with each other.
- Outliers were observed in passenger fare values.

8. Conclusion

Exploratory Data Analysis helped us understand the Titanic dataset through statistical methods and visualizations.

The analysis identified missing values, distributions, relationships and important survival patterns.

This project improved my understanding of data exploration, data cleaning, visualization and extracting meaningful insights using Python.

Project Outcome: Successfully performed Exploratory Data Analysis on the Titanic dataset using Python libraries.